

Koopman-lifted multi-step predictor: what it does and why it is correct

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Summary. The production predictor is a direct multi-step fit on the 47-feature lift: one independent ridge fit per consumer and horizon, no rollout. On a held-out operating day the delivered-heat forecast sits on the parity diagonal at every horizon, per-consumer R^2 0.987 to 0.999 up to 4 h ahead (Section 3). On the hard held-out PRBS days the one-step suite R^2 (mean over five consumers) is 0.712. The direct fit matches or beats persistence at every horizon in the suite mean (one consumer loses to it at the shortest horizons, where one step ahead is nearly a persistence problem) and pulls ahead of the iterated rollout as the horizon grows (12.1 % ahead at $h = 16$). All five validator invariants pass (Section 4).

1 What this component does and what I did

The open-loop identification stage. It turns the AROMA plant into a linear predictor of each consumer’s delivered heat, 15 min to 4 h ahead ($h = 1, \dots, 16$ at $T_s = 900$ s). The applied control sequence and the demand forecast are the known inputs.

Linear read-out. The lifted state $z_k = \psi(\xi_k)$ is 47 fixed functions of the plant state (36 base features plus 11 exergy bilinears; `koopman/candidate_library.m`). Twelve of the base features carry the past: the source-outlet temperature at $t-1$ and $t-2$ and ten delayed bilinears $T^{(F0)}(t-d) q^{(i)}(t)$. Only the source-outlet channel is delayed, not the full state, so the embedding gives z_k a memory of the source history rather than a complete copy of ξ_{k-1}, ξ_{k-2} . That is the channel the transport delay acts on, which is why it is the one worth lagging.

The lift is a function of the state only. Every one of the 47 features is a function of ξ_k . The source-supply *command* $T^{(0,s)}$ and the demand d never enter ψ ; they enter the predictor through the input block alone. What the lift carries is $T^{(F0)}$, the source-*outlet* state, which is the plant’s response to the command and differs from it by up to 15.7 K on the identification data. This is the point the midterm review raised, and `candidate_library.m` is written to respect it.

One channel needs saying plainly. $T^{(i,r)}$, the consumer return temperature, appears both inside z (through $\theta^{(i)}$ and its own feature) and inside the input u . In the formulation the substation return is a control input, so the QP carries it as a decision variable. The plant here does not track that command: the substations set their own returns from the energy balance, and `kpc_step_loop.m` applies only $T^{(0,s)}$ and the flow references. In the identification data the $T^{(i,r)}$ input channel is therefore the realised return, a plant state. I have left it in the input stack because the formulation puts it there, but the honest reading is that it is a predictor degree of freedom rather than an actuator, and the closed-loop results do not depend on it being one.

Delivered heat is an exact linear read-out of the lift:

$$c_k^{(i)} = c_p \theta_k^{(i)}, \quad \theta^{(i)} = (T^{(i,s)} - T^{(i,r)}) q^{(i)}. \quad (1)$$

This is the substation energy balance, so it holds exactly. With θ inside z , heat predictions and constraints are linear in z , so the MPC is a convex QP.

Predictor. V_{seq} is direct multi-step: one independent linear fit per (consumer i , horizon h),

$$\theta_{k+h}^{(i)} = a_{h,i}^\top z_k + b_{h,i}^\top V_k^{(h)}, \quad V_k^{(h)} = [u_k; \dots; u_{k+h-1}; d_{k+1}; \dots; d_{k+h}], \quad (2)$$

with $u = [T^{(0,s)}; r_q; T^{(i,r)}]$ and d the demand forecast; the fitted $a_{h,i}, b_{h,i}$ are stored as `fits.cp`, `fits.dp` (`koopman/fit_vseq.m`). No fit feeds another, so one-step error cannot compound. Each fit is a split ridge,

$$\min_{a,b} \sum_k (\theta_{k+h}^{(i)} - a^\top z_k - b^\top V_k^{(h)})^2 + \lambda_z \|a\|^2 + \lambda_v \|b\|^2, \quad (3)$$

weighted per trajectory (the operational days carry weight 8 against 1 for the PRBS days, `p.o1.op_weight`, so the fit leans toward the regime the controller runs in; both predictors are fitted under the same weighting, so the comparison between them stays fair). It is a separate penalty on the lift block a and the input/demand block b , with (λ_z, λ_v) picked per (h, i) on validation RMSE.

Baselines. A zero-order-hold (persistence) and the standard iterated one-step form (EDMDc): a single linear model $z_{k+1} = Az_k + B[u_k; d_{k+1}]$ fitted by ridge on the same lift and data as the direct maps,

$$[A \ B] = Z_+ X^\top (XX^\top + \lambda I)^{-1}, \quad X = [z_k; u_k; d_{k+1}], \quad (4)$$

Z_+ the next-step lifts, λ picked on the one-step validation RMSE. A multi-step forecast rolls this one model forward,

$$z_{k+h} = A^h z_k + \sum_{j=0}^{h-1} A^{h-1-j} B[u_{k+j}; d_{k+j+1}], \quad \theta_{k+h}^{(i)} = C_\theta z_{k+h}, \quad (5)$$

so a one-step error is fed back in and compounds through A^h , which the per-horizon direct fit (2) avoids by construction. The shipped fit has $\rho(A) = 0.9975 < 1$, so the rollout stays bounded and the comparison isolates predictor structure (`koopman/fit_iterated.m`).

predictor	structure
ZOH	$\theta_{k+h}^{(i)} = \theta_k^{(i)}$ (persistence)
iterated (A, B)	$z_{k+1} = Az_k + B[u_k; d_{k+1}]$, applied h times
V_{seq} (this work)	eq. (2), one independent fit per (h, i)

Table 1: The three predictors. All see the same lift, data and demand.

Data. 20 training, 2 validation and 3 test PRBS days (24h each), where PRBS is a pseudo-random binary excitation of the inputs. Flow bounds are $[0.3, 1.5] \times \text{nominal}$, supply-temperature amplitude 6K, dwell times pairwise-coprime. PRBS spends much of its time capacity-limited, which dilutes the demand coefficient, so the fit adds 8 smooth operating days at `op_weight` = 8 to calibrate the deployment regime (same weighting for both predictors). One honest limitation: the slowest transport paths settle in several hours, longer than the longest dwell. Six extra slow-dwell days cut open-loop long-horizon error by about a quarter but came out 0.3pp worse in a paired closed-loop day, so the set was kept as is and slow-band excitation is future work.

Code layout

file	role
koopman/candidate_library.m	the 47-feature lift, used everywhere
koopman/fit_vseq.m	per-(consumer, horizon) ridge fit of the direct maps
koopman/fit_iterated.m	the one-step (A, B) baseline, same lift and data
check_excitation.m, check_conditioning.m	data adequacy checks (in <code>koopman/</code>)
demo_predictor.m	test evaluation, figures, <code>demo.mat</code>
tests/validate_predictor.m	the coded invariants of Section 4

2 How to reproduce

From the repository root in MATLAB, `demo_predictor` loads the shipped fits and the three held-out test days, prints the per-horizon suite RMSE for ZOH, iterated and V_{seq} , writes the figures, and runs the invariants inline (~ 30 – 60 s, refits nothing). `tests/validate_predictor` re-checks the invariants against `demo.mat`. See README. Expected console output:

```
provenance ok: demo.mat matches the fits (n.z = 47, 16 horizons)
T1 ok: V_seq RMSE <= ZOH RMSE at every horizon
T2 ok: V_seq beats iterated at h = 16 by 12.1 % RMSE (suite mean)
T3 ok: V_seq 1-step suite R^2 = 0.712
T3 floor ok: weakest consumer 1-step R^2 = 0.560
T4 ok: worst-consumer NRMSE at h = 1 = 19.1 %
T5 ok:  $q_{k+1} = a q_k + (1 - a) r_{q,k}$  holds (max residual 0.00e+00 kg/s)
```

3 The figures

Both figures below are scored on a *held-out* operating day. The operational days in the training set sit on a 3 h phase grid ($t_{\text{off}} = 0, 3, \dots, 21$ h; `data/generate_operational.m`), so I score at 10.5 h, half way between two of them. The day is the deployment regime but not one the fit has seen.

3.1 Multi-horizon parity, held-out day (viz_predictor_parity.pdf)

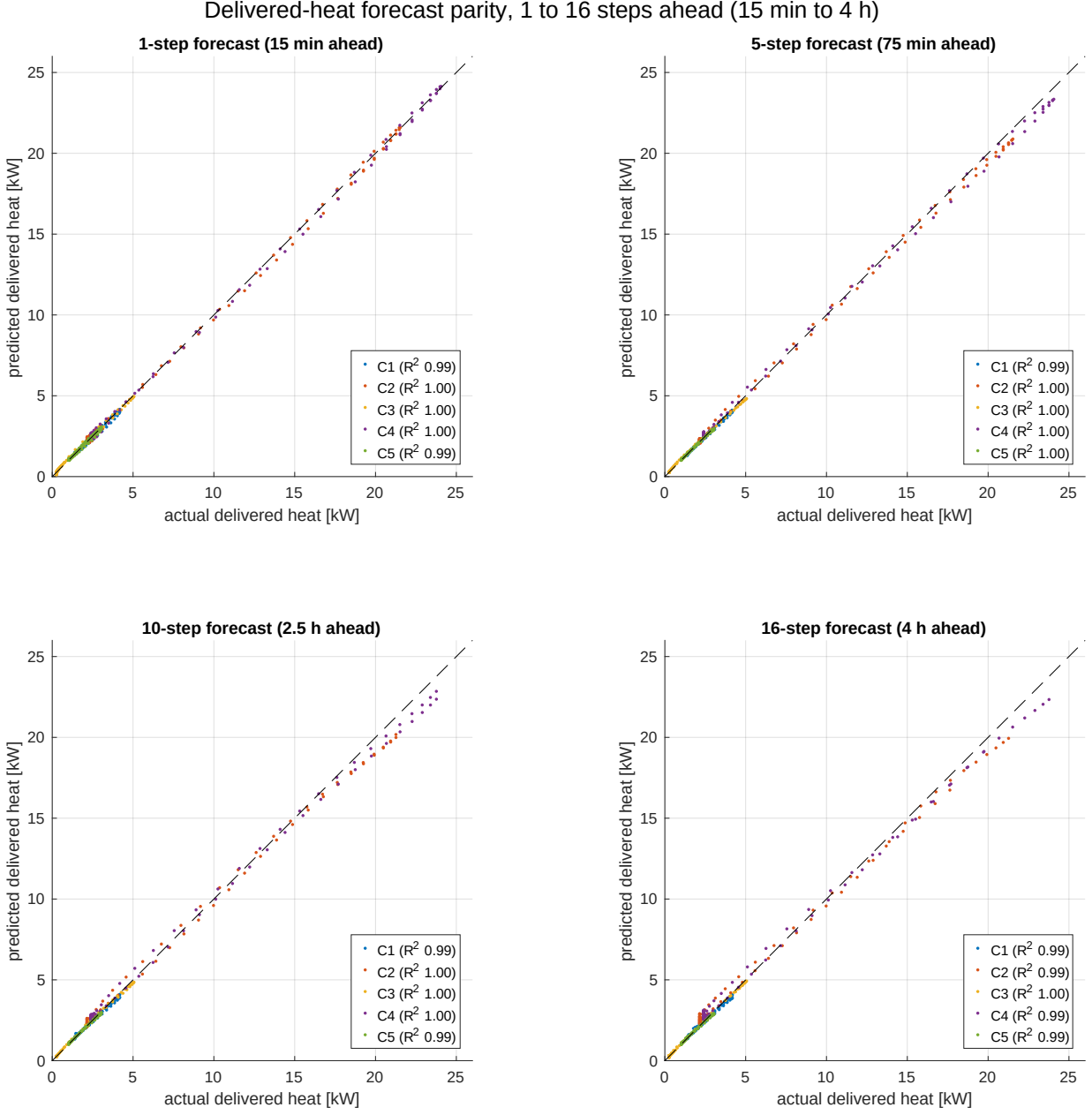


Figure 1: Delivered-heat parity on a held-out operating day at 1, 5, 10 and 16 steps ahead: every consumer on the diagonal at every horizon, per-panel R^2 0.987 to 0.999. The only departure is slight peak-load attenuation at 10 and 16 steps, costing about 0.01 in R^2 .

Deployment-regime result. The held-out PRBS gates (Section 3.3) guard against over-fitting the smooth days.

3.2 Forecast skill versus horizon, held-out day (viz_predictor_error.pdf)

The headline open-loop result. From $h = 5$ on the direct map has roughly three times lower RMSE. The curves coincide at $h = 1$ (identical information there), so the widening gap is pure structure.

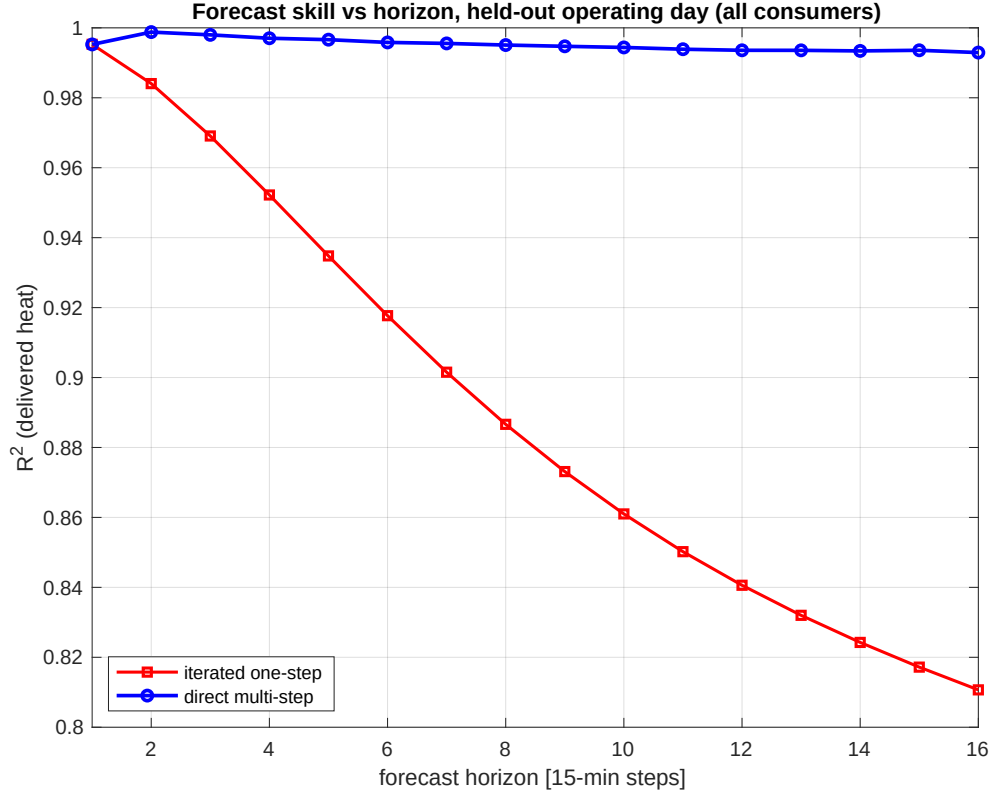


Figure 2: Delivered-heat R^2 versus horizon on a held-out operating day: the direct predictor stays near $R^2 = 1$ to 4 h (0.993 at $h = 16$); the iterated one-step starts equal and decays to 0.811 as its rollout compounds.

3.3 NRMSE versus horizon, held-out PRBS (rmse_horizon.pdf)

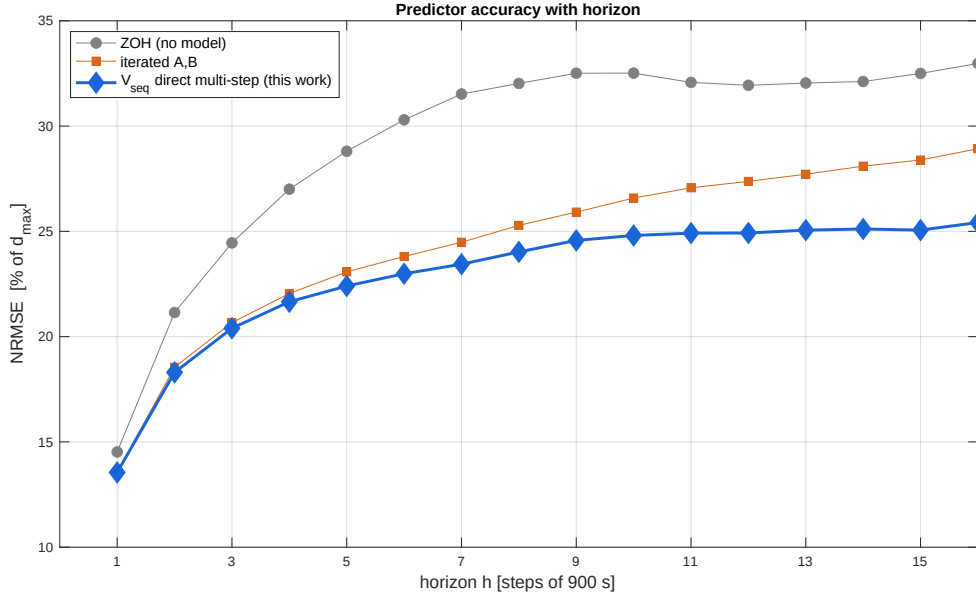


Figure 3: Suite NRMSE versus horizon for ZOH, iterated (A, B) and V_{seq} on the PRBS test set: V_{seq} is lowest at every horizon and the gap to the rollout widens with h .

The stress regime: three held-out broadband PRBS days. At $h = 16$ the three sit at 32.97, 28.92 and 25.42 % of peak demand for ZOH, iterated and V_{seq} ; the direct fit is about 23 % below persistence. At $h = 1$ the margin over persistence is small: one step ahead at 15 minutes is nearly a persistence problem.

4 The automated checks

`tests/validate_predictor.m` runs a provenance guard ($n_z = 47$, the 16 horizons, a coefficient checksum) and then five invariants. The guard is what ties the two together: the errors themselves are computed by `demo_predictor.m` on the held-out test days and saved, and the validator asserts the invariants on those saved arrays rather than re-running the forecasts, except T5, which it re-evaluates directly. So the split is deliberate: the demo does the work, the validator checks it and refuses to run against results that do not match the shipped fits. Labels T1 to T5 number them in print order.

#	invariant	bar	result
T1	$V_{\text{seq}} \text{ RMSE} \leq \text{ZOH RMSE}$	every $h = 1..16$	holds
T2	$V_{\text{seq}} \text{ RMSE} < \text{iterated at } h = 16$	$<$	12.1 % lower
T3	1-step suite R^2 , and weakest consumer	$\geq 0.7, \geq 0.5$	0.712, 0.560
T4	worst-consumer 1-step NRMSE	$\leq 30 \%$	19.1 %
T5	$q_{k+1} = a q_k + (1 - a) r_{q,k}$, every trajectory	residual ≈ 0	0.00e+00 kg/s

Table 2: The validator invariants and their results on the saved data.

T1 pins the structural floor: a fit that loses to persistence is broken. T2 makes the headline claim falsifiable. The direct predictor beating the iterated rollout at the longest horizon is expected for a lift misspecified by construction (Chevillon, 2007), and reverses in the well-specified linear-AR case (Marcellino, Stock and Watson, 2006). T3 and T4 bound absolute accuracy on unseen data; the per-consumer floor stops a good suite mean from hiding one bad consumer. T5 verifies the plant-side flow identity at machine precision. The bars are usable-not-optimal floors: $R^2 \geq 0.7$ is a loose conventional threshold (cleared modestly on PRBS, while the deployment regime reaches 0.99), 30 % NRMSE is an engineering ceiling, and T5 is an exact identity.

Two upstream checks assert the data itself is adequate. `check_excitation.m` confirms the PRBS set is persistently exciting. `check_conditioning.m` reports that the lift is rank deficient by exactly six: $\text{rank}(Z) = 41 = 47 - 6$. The six are not numerical noise, they are exact linear identities I can name ($Q_{\text{net}} = \sum_i q^{(i)}$, and the five per-consumer exergy-difference identities). So the unregularised normal equations are singular, and the ridge is what makes the fit well posed: it absorbs those six null directions rather than fitting them. The lift is therefore redundant by construction and not ill conditioned by accident, and the λ_z penalty is doing structural work, not just shrinkage.